

Lecture 9

Lecturer: Jai Moondra

Scribe: Jai & Claude

1 Introduction

1.1 The problem

Let N be a finite ground set, $|N| = n$. A set function $f : 2^N \rightarrow \mathbb{R}$ is *submodular* if

$$f(A) + f(B) \geq f(A \cup B) + f(A \cap B) \quad \text{for all } A, B \subseteq N,$$

equivalently if marginal values decrease: writing $f_A(i) = f(A \cup \{i\}) - f(A)$, we have $f_A(i) \geq f_B(i)$ whenever $A \subseteq B$ and $i \notin B$. It is *monotone* if $f(A) \leq f(B)$ for $A \subseteq B$, and *modular* if the defining inequality is an equality, in which case $f(A) = f(\emptyset) + \sum_{i \in A} w_i$ for some weights w . Throughout, f is non-negative and is given by a value oracle.

A family $\mathcal{I} \subseteq 2^N$ is *downward closed* (an *independence system*) if $A \subseteq B \in \mathcal{I}$ implies $A \in \mathcal{I}$. The problem is

$$\text{OPT} = \max_{S \in \mathcal{I}} f(S).$$

Examples of $\mathcal{I}$: the independent sets of a matroid, or of an intersection of k matroids; the sets satisfying a knapsack constraint $\sum_{i \in S} a_i \leq B$, or several of them; the routable sets of an unsplittable-flow instance; and intersections of any of these. Examples of f : cut functions (non-monotone), coverage functions (monotone), rank functions of matroids (monotone), and linear functions. Even unconstrained maximization contains Max-Cut, and cardinality-constrained coverage is Max- k -Cover, so we look for approximation algorithms.

1.2 The multilinear extension

The *multilinear extension* of f is $F : [0, 1]^N \rightarrow \mathbb{R}$,

$$F(x) = \sum_{S \subseteq N} f(S) \prod_{i \in S} x_i \prod_{i \notin S} (1 - x_i) = \mathbb{E}[f(R(x))],$$

where $R(x)$ is the random set containing each i independently with probability x_i . F agrees with f on $\{0, 1\}^N$, is linear in each coordinate separately, and is linear when f is modular. Two properties we will use:

Lemma 1.1. For every $z \in [0, 1]^N$ and $i \neq j$,

$$\frac{\partial F}{\partial x_i}(z) = \mathbb{E}[f_{R(z) \setminus \{i\}}(i)], \quad \frac{\partial^2 F}{\partial x_i \partial x_j}(z) \leq 0, \quad \frac{\partial^2 F}{\partial x_i^2}(z) = 0.$$

Proof. By multilinearity, $\partial F / \partial x_i(z) = F(z|_{z_i=1}) - F(z|_{z_i=0}) = \mathbb{E}[f(R' \cup \{i\}) - f(R')]$ with $R' = R(z) \setminus \{i\}$, which is the first formula; the third is immediate from linearity in x_i . For the second, with $R'' = R(z) \setminus \{i, j\}$,

$$\frac{\partial^2 F}{\partial x_i \partial x_j}(z) = \mathbb{E}[f(R'' \cup \{i, j\}) - f(R'' \cup \{i\}) - f(R'' \cup \{j\}) + f(R'')] \leq 0$$

by submodularity. $\square$

Corollary 1.2. *F is concave along every line whose direction vector is non-negative. In particular $F(bx) \geq bF(x)$ for all $x \in [0, 1]^N$ and $b \in [0, 1]$.*

Proof. For $d \geq 0$, $t \mapsto F(x + td)$ has second derivative $d^\top \nabla^2 F d \leq 0$ by Lemma 1.1. Applying this to $g(t) = F(tx)$ on $[0, 1]$, concavity gives $g(b) \geq (1 - b)g(0) + bg(1) \geq bF(x)$, since $g(0) = f(\emptyset) \geq 0$. $\square$

The inequality $F(bx) \geq bF(x)$ is in general strict: F is not linear along rays. (Take $f(S) = \min(|S|, 1)$ on two elements and $x = (1, 1)$; then $F(x) = 1$ but $F(x/2) = 3/4$.)

1.3 Relaxation and continuous optimization

Let $P_{\mathcal{I}} \subseteq [0, 1]^N$ be a polytope with $\{\mathbf{1}_S : S \in \mathcal{I}\} \subseteq P_{\mathcal{I}}$. We call $P_{\mathcal{I}}$ *solvable* if it has a polynomial-time separation oracle, and *down-monotone* if $0 \leq y \leq x \in P_{\mathcal{I}}$ implies $y \in P_{\mathcal{I}}$. Then

$$\max\{F(x) : x \in P_{\mathcal{I}}\} \geq \max\{f(S) : \mathbf{1}_S \in P_{\mathcal{I}}\} \geq \text{OPT},$$

and the natural plan is to (approximately) maximize F over $P_{\mathcal{I}}$ and then round. F is neither convex nor concave, so neither step is automatic. For the first step we take the following as given.

Assumption 1.3 (Continuous approximation; Theorem 1.1 of the paper). *There is a constant $\alpha > 0$ and a polynomial-time algorithm which, for every down-monotone solvable polytope $P \subseteq [0, 1]^N$ and every non-negative submodular f , returns $x^* \in P$ with*

$$F(x^*) \geq \alpha \cdot \max\{f(S) : \mathbf{1}_S \in P\}.$$

The paper proves this with $\alpha = 0.325$ (and a simpler algorithm with $\alpha = 0.25$, relative to the fractional optimum). For monotone f the continuous greedy algorithm gives $\alpha = 1 - 1/e$, and for modular f the problem is a linear program, $\alpha = 1$. Down-monotonicity of P cannot be dropped in the non-monotone case.

1.4 Rounding: contention resolution schemes

The obvious way to round $x \in P_{\mathcal{I}}$ while preserving $F(x)$ is to output $R(x)$ itself, since $\mathbb{E}[f(R(x))] = F(x)$. But $R(x)$ is usually not in $\mathcal{I}$. A *contention resolution scheme* repairs $R(x)$ by deleting elements, in a way that no element is deleted too often. Because it is often useful to first scale x down, the definition has two parameters.

Definition 1.4 (CR scheme). Let $b \in (0, 1]$ and $c \in (0, 1]$. A (b, c) -balanced CR scheme for $P_{\mathcal{I}}$ is a (possibly randomized) procedure π which, for every $x \in P_{\mathcal{I}}$, takes as input the random set $R = R(bx)$ and outputs $I = \pi(R)$ with

- (i) $I \subseteq R$ and $I \in \mathcal{I}$ (always), and
- (ii) $\mathbb{P}[i \in I \mid i \in R] \geq c$ for every $i \in N$,

where the probability is over both R and the internal randomness of π . A $(1, c)$ -balanced scheme is called c -balanced. The scheme is

- monotone if $\mathbb{P}[i \in I \mid R = R_1] \geq \mathbb{P}[i \in I \mid R = R_2]$ whenever $i \in R_1 \subseteq R_2$ (an element is less likely to survive when more competitors are sampled);

- strict if $\mathbb{P}[i \in I \mid i \in R] = c$ exactly, for every i .

Some comments on the definition.

- The scheme is defined relative to a particular polytope $P_{\mathcal{I}}$ and must work for every $x \in P_{\mathcal{I}}$. The alteration $R \mapsto \pi(R)$ is allowed to depend on x ; for matroids it will.
- Property (ii) is a statement about marginals only. A modular f is then handled by linearity of expectation:

$$\mathbb{E}[f(I)] - f(\emptyset) = \sum_i w_i \mathbb{P}[i \in I] \geq c \sum_i w_i b x_i = c(F(bx) - f(\emptyset)),$$

with no further assumption on π . For submodular f this argument fails, because $f(I)$ is not a sum of per-element terms, and we will need monotonicity of π .

- Monotonicity is not automatic. The natural greedy alteration “scan R in a fixed order and keep i if it is still feasible” is not monotone in general: adding an element to R can knock out a competitor of i and thereby let i in.
- As $b \rightarrow 0$ we expect $c \rightarrow 1$: a sparser sample has less contention. The (b, c) families we construct have this property, and it is what makes schemes for different constraints composable.
- Strictness is needed, in the paper, for non-monotone f ; we will show below that a simple pruning step makes it unnecessary.

1.5 An easy scheme for uniform matroids

Let $\mathcal{M}$ be the uniform matroid of rank k on N , i.e. $\mathcal{I}_{\mathcal{M}} = \{S : |S| \leq k\}$, whose matroid polytope is $P(\mathcal{M}) = \{x \in [0, 1]^N : \sum_i x_i \leq k\}$. The following scheme is simple to state and to analyze, at the price of a worse constant than the optimal $\frac{1-e^{-b}}{b}$.

Random-subset scheme. Given $x \in P(\mathcal{M})$ and $b \in (0, 1]$, sample $R = R(bx)$. If $|R| \leq k$, output $I = R$; otherwise output a uniformly random k -element subset $I \subseteq R$.

Theorem 1.5. *The random-subset scheme is a monotone $(b, \frac{1}{1+b})$ -balanced CR scheme for $P(\mathcal{M})$.*

Proof. Feasibility is immediate. For monotonicity, given R every element of R survives with the same probability,

$$\mathbb{P}[i \in I \mid R] = \min\left(1, \frac{k}{|R|}\right) \quad (i \in R),$$

which is non-increasing in R .

For balance, fix i and condition on $i \in R$. Let $X = |R \setminus \{i\}|$, a sum of independent Bernoulli variables with

$$\mathbb{E}[X] = b \sum_{j \neq i} x_j \leq bk$$

by the constraint defining $P(\mathcal{M})$. Then $\mathbb{P}[i \in I \mid i \in R] = \mathbb{E}\left[\min\left(1, \frac{k}{1+X}\right)\right]$. The function $t \mapsto \min\left(1, \frac{k}{1+t}\right)$ is not convex on $t \geq 0$ (it has a concave kink at $t = k - 1$), so we compare it with a convex minorant: for all $t \geq 0$,

$$\min\left(1, \frac{k}{1+t}\right) \geq \frac{k}{k+t}.$$

Indeed, if $t \leq k - 1$ the left side is $1 \geq \frac{k}{k+t}$, and if $t \geq k - 1$ the left side is $\frac{k}{1+t} \geq \frac{k}{k+t}$ since $k \geq 1$. The function $t \mapsto \frac{k}{k+t}$ is convex, so by Jensen's inequality

$$\mathbb{P}[i \in I \mid i \in R] \geq \mathbb{E}\left[\frac{k}{k+X}\right] \geq \frac{k}{k+\mathbb{E}[X]} \geq \frac{k}{k+bk} = \frac{1}{1+b}.$$

□

Remark. (1) The scheme is not strict: $\mathbb{P}[i \in I \mid i \in R]$ depends on x through $\mathbb{E}[X]$. The pruning step of Theorem 2.3 makes this irrelevant.

(2) The bound $\frac{1}{1+b}$ is tight for this scheme. Take $k = 1$, $b = 1$, and $x = (\varepsilon, 1 - \varepsilon)$ on two elements. Element 1 survives with probability $\varepsilon \cdot 1 + (1 - \varepsilon) \cdot \frac{1}{2} \rightarrow \frac{1}{2}$. Compare with the optimal $\left(b, \frac{1-e^{-b}}{b}\right)$: at $b = 1$ this is $1 - 1/e \approx 0.632$ versus $\frac{1}{2}$; as $b \rightarrow 0$ the two behave as $1 - \frac{b}{2}$ versus $1 - b$.

(3) The same example shows why an optimal scheme must depend on x beyond the sampling step. Element 1 is rarely sampled, but whenever it is, it almost always finds element 2 present and loses a fair coin flip. To guarantee survival probability $1 - 1/e$ to every element, the resolution must favor element 1 in that contention, i.e. the alteration probabilities must be tailored to the marginals. This is what the LP of Section 4 does for general matroids.

(4) A still simpler scheme, “output R if $|R| \leq k$ and $\emptyset$ otherwise”, is monotone and $(b, 1 - b)$ -balanced by Markov's inequality ($\mathbb{P}[X \geq k] \leq \mathbb{E}[X]/k \leq b$), but strictly worse: it discards everything on overflow instead of salvaging k elements.

1.6 The framework

Given Assumption 1.3 and a monotone (b, c) -balanced CR scheme for $P_{\mathcal{I}}$, the algorithm is:

1. Compute $x^* \in P_{\mathcal{I}}$ with $F(x^*) \geq \alpha \text{OPT}$.
2. Sample $R = R(bx^*)$, run the scheme to get $I \in \mathcal{I}$, and output the pruning $I' \subseteq I$ defined in Section 2.

Section 2 shows that $\mathbb{E}[f(I')] \geq bcF(x^*)$, hence this is an αbc -approximation, and that schemes for several constraints can be composed. Section 3 applies this to intersections of k matroids, assuming a $\left(b, \frac{1-e^{-b}}{b}\right)$ -balanced scheme for a single matroid, and obtains $\Omega(1/k)$. Section 4 constructs the single-matroid scheme.

2 Rounding through a CR scheme

We use the following classical inequality; a proof is in the appendix.

Lemma 2.1 (FKG / Harris inequality, product measures). *Let R be a random subset of a finite set in which elements are included independently, and let φ, ψ be real functions of R that are both non-increasing (or both non-decreasing) under inclusion. Then*

$$\mathbb{E}[\varphi(R)\psi(R)] \geq \mathbb{E}[\varphi(R)]\mathbb{E}[\psi(R)].$$

Fix once and for all an ordering $1, 2, \dots, n$ of N ; write $[i] = \{1, \dots, i\}$.

Definition 2.2 (Pruning). Given $I \subseteq N$, its pruning $I' \subseteq I$ is obtained by scanning $i = 1, \dots, n$ and keeping $i \in I$ iff $f_{I' \cap [i-1]}(i) > 0$, where $I' \cap [i-1]$ is the set kept so far.

Pruning only deletes, so $I' \in \mathcal{I}$ whenever $I \in \mathcal{I}$. It uses value queries to f , so the combined procedure is no longer oblivious to f ; this is harmless for our purposes.

Theorem 2.3. *Let $x \in P_{\mathcal{I}}$, let I be the output of a monotone (b, c) -balanced CR scheme on x , and let I' be its pruning. Then*

$$\mathbb{E}[f(I')] \geq cF(bx) \geq bcF(x).$$

If f is monotone, the same holds for I itself. If the scheme is strict, the same holds for I itself for every non-negative submodular f .

Proof. The second inequality is Corollary 1.2. For the first, write $R = R(bx)$ and define, for each i ,

$$\varphi_i(R) = f_{R \cap [i-1]}(i), \quad \varphi_i^+ = \max(\varphi_i, 0), \quad \psi_i(R) = \mathbb{P}[i \in I \mid R].$$

$\psi_i(R)$ is the probability, over the scheme's internal randomness alone, that i survives when the sample is R ; it is 0 when $i \notin R$.

Step 1: $f(I') \geq f(\emptyset) + \sum_i \mathbf{1}[i \in I] \varphi_i^+(R)$ *pointwise.* Telescoping along the ordering,

$$f(I') = f(\emptyset) + \sum_{i \in I'} f_{I' \cap [i-1]}(i).$$

Consider $i \in I$. If $i \in I'$, its term is $f_{I' \cap [i-1]}(i)$, which is > 0 by the pruning rule and $\geq \varphi_i(R)$ by submodularity, since $I' \cap [i-1] \subseteq R \cap [i-1]$; hence it is $\geq \varphi_i^+(R)$. If $i \in I \setminus I'$, it contributes nothing on the left, and $f_{I' \cap [i-1]}(i) \leq 0$ by the pruning rule, so by submodularity $\varphi_i(R) \leq f_{I' \cap [i-1]}(i) \leq 0$ and $\varphi_i^+(R) = 0$ on the right. Elements $i \notin I$ contribute nothing on either side.

Step 2: for each i , $\mathbb{E}[\mathbf{1}[i \in I] \varphi_i^+(R)] \geq c \mathbb{E}[\mathbf{1}[i \in R] \varphi_i(R)]$. Conditioning on R (which removes the scheme's randomness) and then on the event $i \in R$,

$$\mathbb{E}[\mathbf{1}[i \in I] \varphi_i^+(R)] = \mathbb{E}[\psi_i \varphi_i^+] = \mathbb{P}[i \in R] \cdot \mathbb{E}[\psi_i \varphi_i^+ \mid i \in R].$$

Given $i \in R$, the other coordinates of R remain independent, so Lemma 2.1 applies to functions of $R \setminus \{i\}$. Both φ_i^+ and ψ_i are non-increasing there: φ_i^+ by submodularity (and because $\max(\cdot, 0)$ preserves monotonicity), and ψ_i by monotonicity of the scheme. Hence

$$\mathbb{E}[\psi_i \varphi_i^+ \mid i \in R] \geq \mathbb{E}[\psi_i \mid i \in R] \cdot \mathbb{E}[\varphi_i^+ \mid i \in R] \geq c \mathbb{E}[\varphi_i^+ \mid i \in R] \geq c \mathbb{E}[\varphi_i \mid i \in R],$$

where the middle inequality uses $\varphi_i^+ \geq 0$ together with property (ii) of the scheme. Multiplying by $\mathbb{P}[i \in R]$ gives the claim.

Step 3. Take expectations in Step 1, apply Step 2 termwise, and use the exact telescoping $f(R) = f(\emptyset) + \sum_i \mathbf{1}[i \in R] \varphi_i(R)$:

$$\mathbb{E}[f(I')] \geq f(\emptyset) + c(\mathbb{E}[f(R)] - f(\emptyset)) = (1 - c)f(\emptyset) + cF(bx) \geq cF(bx),$$

using $f \geq 0$ and $c \leq 1$.

The variants. If f is monotone, $\varphi_i \geq 0$ already, so Step 1 holds with I in place of I' and φ_i in place of φ_i^+ , and Steps 2–3 go through unchanged. If the scheme is strict and f is arbitrary, run Steps 1–3 with I and φ_i : in Step 2 the FKG inequality gives

$$\mathbb{E}[\psi_i \varphi_i \mid i \in R] \geq \mathbb{E}[\psi_i \mid i \in R] \mathbb{E}[\varphi_i \mid i \in R] = c \mathbb{E}[\varphi_i \mid i \in R]$$

with equality in the last step by strictness, so no sign information about φ_i is needed. (Without strictness the factor $\mathbb{E}[\psi_i \mid i \in R]$ could exceed c , which hurts when $\mathbb{E}[\varphi_i \mid i \in R] < 0$; pruning is what removes this obstacle.) $\square$

Corollary 2.4. *Under Assumption 1.3, if $P_{\mathcal{I}}$ admits a polynomial-time monotone (b, c) -balanced CR scheme, the framework of Section 1.5 is a randomized αbc -approximation for $\max_{S \in \mathcal{I}} f(S)$.*

Proof. $\mathbb{E}[f(I')] \geq bcF(x^*) \geq \alpha bc \text{ OPT}$. $\square$

2.1 Composition

Theorem 2.5 (Composition). *Let $\mathcal{I} = \bigcap_{j=1}^k \mathcal{I}_j$ and $P = \bigcap_{j=1}^k P_{\mathcal{I}_j}$. If each $P_{\mathcal{I}_j}$ admits a monotone (b, c_j) -balanced CR scheme, then P admits a monotone $(b, \prod_j c_j)$ -balanced CR scheme for $\mathcal{I}$. If each scheme runs in polynomial time, so does the composed one. In particular, if $c_j = c$ for all j , P has a monotone (b, c^k) -balanced scheme.*

Proof. Given $x \in P$, note $x \in P_{\mathcal{I}_j}$ for each j . Sample $R = R(bx)$ once. Run each scheme j on this same R , with independent internal randomness, obtaining $I_j \in \mathcal{I}_j$; output $I = \bigcap_j I_j$. Then $I \subseteq R$ and $I \in \mathcal{I}$, giving property (i).

For property (ii), let $\psi_j(R) = \mathbb{P}[i \in I_j \mid R]$. Given R the k runs are independent, so $\mathbb{P}[i \in I \mid R] = \prod_j \psi_j(R)$. Each ψ_j is non-negative and, by monotonicity of scheme j , non-increasing on sets containing i ; hence every partial product $\prod_{j \leq m} \psi_j$ is non-negative and non-increasing. Conditioning on $i \in R$ (a product distribution on the other coordinates) and applying Lemma 2.1 $k - 1$ times,

$$\mathbb{P}[i \in I \mid i \in R] = \mathbb{E} \left[\prod_j \psi_j \mid i \in R \right] \geq \prod_j \mathbb{E}[\psi_j \mid i \in R] \geq \prod_j c_j.$$

Monotonicity of the composed scheme: $\mathbb{P}[i \in I \mid R] = \prod_j \psi_j(R)$ is a product of non-negative non-increasing functions, hence non-increasing. $\square$

Remark. Monotonicity of the individual schemes is used twice in this section: in Theorem 2.3, to relate $\mathbb{E}[f(I)]$ to $F(bx)$, and in Theorem 2.5, to make the survival probabilities multiply rather than merely be independent given R . The composed relaxation is the intersection of the individual relaxations, which is still solvable, so Assumption 1.3 applies to it.

3 Intersections of k matroids

Let $\mathcal{M}_1, \dots, \mathcal{M}_k$ be matroids on N with independent-set families $\mathcal{I}_j$ and rank functions r_j , and let $\mathcal{I} = \bigcap_j \mathcal{I}_j$. For a matroid $\mathcal{M}$ with rank function r , the *matroid polytope* is

$$P(\mathcal{M}) = \{x \in \mathbb{R}_{\geq 0}^N : x(S) \leq r(S) \ \forall S \subseteq N\}.$$

Fact 3.1 (Edmonds). *$P(\mathcal{M})$ is the convex hull of $\{\mathbf{1}_S : S \in \mathcal{I}(\mathcal{M})\}$, and it has a polynomial-time separation oracle. The greedy algorithm returns a maximum-weight independent set for any non-negative weights.*

We take

$$P = \bigcap_j P(\mathcal{M}_j)$$

as the relaxation of $\mathcal{I}$. It is down-monotone (each $P(\mathcal{M}_j)$ is defined by $x \geq 0$ and upper bounds on sums of coordinates), solvable (run the k separation oracles), and contains $\mathbf{1}_S$ for every $S \in \mathcal{I}$.

The single-matroid scheme is the subject of Section 4; we state it here.

Theorem 3.2. *For every matroid $\mathcal{M}$ and every $b \in (0, 1]$, $P(\mathcal{M})$ admits a monotone $(b, \frac{1-e^{-b}}{b})$ -balanced CR scheme. It can be constructed and run in time polynomial in n and $1/\varepsilon$ at the cost of replacing c by $c - \varepsilon$ (this last statement is Theorem 4.6 of the paper, which we do not prove).*

Note $\frac{1-e^{-b}}{b}$ is $1 - 1/e$ at $b = 1$ and tends to 1 as $b \rightarrow 0$, while $b \cdot \frac{1-e^{-b}}{b} = 1 - e^{-b}$ is increasing in b . So for a single matroid one would take $b = 1$; the scaling is useful only when composing.

Theorem 3.3. *Under Assumption 1.3 and Theorem 3.2, there is a randomized polynomial-time algorithm returning $S \in \mathcal{I}$ with*

$$\mathbb{E}[f(S)] \geq \frac{2\alpha}{e(k+1)} \text{OPT} \geq \frac{\alpha}{ek} \text{OPT} \quad (\text{up to an additive } \varepsilon \text{ in the constant}).$$

Proof. Set $b = \frac{2}{k+1}$. Compose the k schemes of Theorem 3.2 via Theorem 2.5: P admits a monotone (b, c) -balanced scheme with

$$c = \left(\frac{1 - e^{-b}}{b} \right)^k.$$

By Corollary 2.4 the framework is an abc -approximation. It remains to bound bc from below.

Claim: $\frac{1-e^{-s}}{s} \geq 1 - \frac{s}{2}$ for $s > 0$. Let $\phi(s) = 1 - s + \frac{s^2}{2} - e^{-s}$; then $\phi(0) = 0$ and $\phi'(s) = -1 + s + e^{-s} \geq 0$ because $e^{-s} \geq 1 - s$. So $\phi \geq 0$, i.e. $1 - e^{-s} \geq s - \frac{s^2}{2}$; divide by s .

With $s = b = \frac{2}{k+1}$ the claim gives $1 - \frac{b}{2} = \frac{k}{k+1}$, hence

$$c \geq \left(\frac{k}{k+1} \right)^k = \left(1 + \frac{1}{k} \right)^{-k} \geq \frac{1}{e},$$

using $k \ln(1 + \frac{1}{k}) \leq 1$. Therefore $bc \geq \frac{2}{e(k+1)}$, and $\frac{2}{k+1} \geq \frac{1}{k}$ for $k \geq 1$. $\square$

4 A CR scheme for one matroid

Fix a matroid $\mathcal{M} = (N, \mathcal{I}_{\mathcal{M}})$ with rank function r , a point $x \in P(\mathcal{M})$ and $b \in (0, 1]$. Write $p = bx$ and $R = R(p)$. The goal is to show that a monotone c -balanced scheme exists for R with $c = \frac{1-e^{-b}}{b}$. The strategy: write the best achievable c as a linear program, pass to the dual, and observe that the dual is exactly a correlation-gap statement about weighted rank functions of $\mathcal{M}$, which we then prove.

4.1 The LP and its dual

Let Π be the set of all deterministic CR schemes for the sample R : maps $\pi : 2^N \rightarrow \mathcal{I}_{\mathcal{M}}$ with $\pi(R) \subseteq R$ for all R . This is a finite set. For $\pi \in \Pi$ and $i \in N$ let

$$q_{i,\pi} = \mathbb{P}_R[i \in \pi(R)].$$

A randomized scheme is a probability distribution λ over Π : draw $\pi \sim \lambda$, output $\pi(R)$. Its survival probability for i is $\mathbb{P}[i \in I] = \sum_{\pi} \lambda_{\pi} q_{i,\pi}$, so it is c -balanced iff $\sum_{\pi} \lambda_{\pi} q_{i,\pi} \geq cp_i$ for all i . (Every randomized scheme with finitely many outcomes is of this form; conversely every λ gives a scheme.) The best c is therefore

$$c^* = \max \left\{ c : \sum_{\pi \in \Pi} \lambda_{\pi} q_{i,\pi} \geq cp_i \ \forall i \in N, \quad \sum_{\pi} \lambda_{\pi} = 1, \quad \lambda \geq 0 \right\}. \quad (\text{LP1})$$

Its dual, with a variable $y_i \geq 0$ for each covering constraint and a free variable μ for the normalization, is

$$\mu^* = \min \left\{ \mu : \sum_{i \in N} q_{i,\pi} y_i \leq \mu \ \forall \pi \in \Pi, \quad \sum_i p_i y_i = 1, \quad y \geq 0 \right\}. \quad (\text{DP1})$$

By strong duality $c^* = \mu^*$ (both programs are feasible and bounded). To use this we interpret the dual constraint. For fixed $y \geq 0$,

$$\sum_i q_{i,\pi} y_i = \sum_i y_i \mathbb{P}[i \in \pi(R)] = \mathbb{E}[y(\pi(R))],$$

the expected y -weight of the scheme's output. So (DP1) asks for

$$\mu^* = \min_{\substack{y \geq 0, \\ \sum_i p_i y_i = 1}} \max_{\pi \in \Pi} \mathbb{E}[y(\pi(R))].$$

4.2 Greedy schemes suffice, and are monotone

For $y \geq 0$ define the *weighted rank function*

$$r_y(S) = \max\{y(T) : T \subseteq S, \ T \in \mathcal{I}_{\mathcal{M}}\},$$

and let $\pi_y \in \Pi$ be the *greedy scheme*: $\pi_y(R)$ is the maximum- y -weight independent subset of R computed by the greedy algorithm, scanning R in non-increasing order of y with a fixed tie-breaking rule. By Fact 3.1, $y(\pi_y(R)) = r_y(R)$, and for any $\pi \in \Pi$, $y(\pi(R)) \leq r_y(R)$ since $\pi(R)$ is an independent subset of R . Taking expectations,

$$\mathbb{E}[y(\pi(R))] \leq \mathbb{E}[r_y(R)] = \mathbb{E}[y(\pi_y(R))] \quad \text{for all } \pi \in \Pi. \quad (1)$$

Hence the inner maximum in (DP1) is attained by π_y , and

$$\mu^* = \min_{\substack{y \geq 0, \\ \sum_i p_i y_i = 1}} \mathbb{E}[r_y(R)]. \quad (2)$$

Lemma 4.1 (Greedy schemes are monotone). *For every y , if $i \in R_1 \subseteq R_2$ and $i \in \pi_y(R_2)$, then $i \in \pi_y(R_1)$.*

Proof. Greedy accepts i iff i is not in the span of the elements of R scanned before i (the accepted elements before i have the same span). Let $B_1 \subseteq B_2$ be the elements of R_1 , resp. R_2 , scanned before i . Then $\text{span}(B_1) \subseteq \text{span}(B_2)$, so $i \notin \text{span}(B_2)$ implies $i \notin \text{span}(B_1)$. $\square$

Lemma 4.2. *Let $\Pi' = \{\pi_y : y \geq 0\}$ (a finite set, one scheme per ordering of N). The LP (LP1) restricted to the variables λ_π , $\pi \in \Pi'$, has optimum c^* , and any optimal λ of the restricted LP is a monotone c^* -balanced scheme.*

Proof. By (1), for every y the dual constraint indexed by π_y implies the constraint indexed by any other π . So (DP1) with constraints restricted to Π' has the same feasible region and value μ^* . The restricted (DP1) is the dual of the restricted (LP1), so by strong duality the restricted (LP1) has value $\mu^* = c^*$. A convex combination of the π_y is monotone because each $R \mapsto \mathbf{1}[i \in \pi_y(R)]$ is non-increasing on sets containing i (Lemma 4.1) and

$$\mathbb{P}[i \in I \mid R] = \sum_y \lambda_{\pi_y} \mathbf{1}[i \in \pi_y(R)]$$

is a non-negative combination of such functions. □

By (2) and Lemma 4.2, the whole question reduces to the following statement about weighted rank functions.

Theorem 4.3. *For every $y \geq 0$,*

$$\mathbb{E}[r_y(R(bx))] \geq \frac{1 - e^{-b}}{b} \sum_i p_i y_i, \quad p = bx.$$

Consequently $c^ \geq \frac{1 - e^{-b}}{b}$, and Theorem 3.2 (existence and monotonicity) holds.*

The rest of this section proves Theorem 4.3.

4.3 Weighted rank functions

Lemma 4.4. *r_y is monotone, submodular, and $r_y(\emptyset) = 0$.*

Proof. Order N so that $y_1 \geq y_2 \geq \dots \geq y_n \geq 0$ and put $y_{n+1} = 0$. Greedy scanning S in this order picks exactly $r(S \cap [j])$ elements from $S \cap [j]$, for every j , and by Fact 3.1 its output has weight $r_y(S)$. Hence

$$r_y(S) = \sum_{j=1}^n y_j (r(S \cap [j]) - r(S \cap [j-1])) = \sum_{j=1}^n (y_j - y_{j+1}) r(S \cap [j]).$$

Each $S \mapsto r(S \cap [j])$ is monotone submodular (a matroid rank function restricted to $[j]$) and the coefficients $y_j - y_{j+1}$ are non-negative. □

Lemma 4.5. *Let $x \in P(\mathcal{M})$ and write $x = \sum_S \alpha_S \mathbf{1}_S$ as a convex combination of independent sets (Fact 3.1). Then $\sum_S \alpha_S r_y(S) = y \cdot x$.*

Proof. For independent S , $r_y(S) = y(S)$. So

$$\sum_S \alpha_S r_y(S) = \sum_S \alpha_S y(S) = y \cdot \sum_S \alpha_S \mathbf{1}_S = y \cdot x.$$

□

4.4 The correlation-gap bound at a scaled point

Lemma 4.6. *Let $g : 2^N \rightarrow \mathbb{R}_+$ be monotone submodular with multilinear extension G , and let $x = \sum_S \alpha_S \mathbf{1}_S$ be any convex combination ($\alpha_S \geq 0$, $\sum_S \alpha_S = 1$). Then for every $t \in [0, 1]$,*

$$G(tx) \geq (1 - e^{-t}) \sum_S \alpha_S g(S).$$

Proof. Let $h(t) = G(tx)$, $g^* = \sum_S \alpha_S g(S)$ and $R = R(tx)$. By Lemma 1.1 (applied to g) and the chain rule,

$$h'(t) = x \cdot \nabla G(tx) = \sum_i x_i \mathbb{E}[g_{R \setminus \{i\}}(i)].$$

Since $R \setminus \{i\} \subseteq R$, submodularity gives $g_{R \setminus \{i\}}(i) \geq g_R(i)$, so

$$h'(t) \geq \sum_i x_i \mathbb{E}[g_R(i)] = \mathbb{E} \left[\sum_S \alpha_S \sum_{i \in S} g_R(i) \right].$$

For fixed R and S , add the elements of $S \setminus R$ to R one at a time; submodularity gives $\sum_{i \in S} g_R(i) \geq g(R \cup S) - g(R)$, and monotonicity gives $g(R \cup S) \geq g(S)$. Averaging over S with weights α_S and taking expectation over R ,

$$h'(t) \geq g^* - h(t).$$

Then

$$\frac{d}{dt}(e^t h(t)) = e^t(h'(t) + h(t)) \geq e^t g^*;$$

integrating from 0 to t gives $e^t h(t) - h(0) \geq (e^t - 1)g^*$, and since $h(0) = g(\emptyset) \geq 0$ we get $h(t) \geq (1 - e^{-t})g^*$. $\square$

At $t = 1$ this is the statement that the correlation gap of monotone submodular functions is $1 - 1/e$; the scaled version is what the parameter b needs.

Proof of Theorem 4.3. Apply Lemma 4.6 to $g = r_y$ (monotone submodular by Lemma 4.4), $t = b$, and the convex combination of x into independent sets given by Fact 3.1. Using Lemma 4.5 for the right-hand side,

$$\mathbb{E}[r_y(R(bx))] = R_y(bx) \geq (1 - e^{-b})y \cdot x = \frac{1 - e^{-b}}{b} y \cdot (bx) = \frac{1 - e^{-b}}{b} \sum_i p_i y_i,$$

where R_y denotes the multilinear extension of r_y . For y with $\sum_i p_i y_i = 1$ this says $\mathbb{E}[r_y(R)] \geq \frac{1 - e^{-b}}{b}$, so by (2) $\mu^* \geq \frac{1 - e^{-b}}{b}$, hence $c^* \geq \frac{1 - e^{-b}}{b}$ by strong duality, and Lemma 4.2 provides a monotone scheme achieving c^* . $\square$

Remark (What is not proved here). (1) The bound is tight: the paper's Theorem 4.9 shows no $(b, \frac{1 - e^{-b}}{b} + \varepsilon)$ -balanced scheme exists even for the uniform matroid of rank one.

(2) The scheme is defined by an LP with exponentially many variables; finding it efficiently (the paper's Theorem 4.6) uses the ellipsoid method on the dual with the greedy scheme as an approximate separation oracle, the quantities q_{i, π_y} being estimated by sampling. This costs an ε and is the source of the $\Omega(1/\text{poly}(n))$ slack in the paper's Theorem 4.7.

(3) The scheme depends on x through the LP, unlike the oblivious alterations used for knapsack and packing constraints.

A Proof of the FKG inequality (Lemma 2.1)

Replacing (φ, ψ) by $(-\varphi, -\psi)$ if necessary, assume both are non-increasing. Identify R with $(X_1, \dots, X_m) \in \{0, 1\}^m$, the X_j independent with $\mathbb{P}[X_j = 1] = p_j$. Induct on m .

Base case $m = 1$. Write $\varphi_s = \varphi(X_1 = s)$, similarly ψ_s , and $p = p_1$. Non-increasing means $\varphi_1 \leq \varphi_0$, $\psi_1 \leq \psi_0$, and

$$\mathbb{E}[\varphi\psi] - \mathbb{E}[\varphi]\mathbb{E}[\psi] = p(1-p)(\varphi_1 - \varphi_0)(\psi_1 - \psi_0) \geq 0.$$

Inductive step. Condition on X_m . For each $s \in \{0, 1\}$ the functions $\varphi(\cdot, s), \psi(\cdot, s)$ of the first $m-1$ coordinates are non-increasing and those coordinates are independent, so by induction

$$\mathbb{E}[\varphi\psi \mid X_m = s] \geq \mathbb{E}[\varphi \mid X_m = s]\mathbb{E}[\psi \mid X_m = s].$$

Next, $\bar{\varphi}(s) := \mathbb{E}[\varphi \mid X_m = s]$ is non-increasing in s : for every fixed value of the first $m-1$ coordinates, $\varphi(\cdot, 1) \leq \varphi(\cdot, 0)$, and averaging over those coordinates (whose law does not depend on s) preserves the inequality; likewise for $\bar{\psi}$. By the base case applied to $\bar{\varphi}, \bar{\psi}$,

$$\mathbb{E}[\varphi\psi] = \mathbb{E}[\mathbb{E}[\varphi\psi \mid X_m]] \geq \mathbb{E}[\bar{\varphi}(X_m)\bar{\psi}(X_m)] \geq \mathbb{E}[\bar{\varphi}(X_m)]\mathbb{E}[\bar{\psi}(X_m)] = \mathbb{E}[\varphi]\mathbb{E}[\psi].$$

References

- [1] Chandra Chekuri, Jan Vondrák, and Rico Zenklusen. Submodular function maximization via the multilinear relaxation and contention resolution schemes. In *Proceedings of the 43rd Annual ACM Symposium on Theory of Computing (STOC)*, pages 783–792, 2011. <https://doi.org/10.1145/1993636.1993740>.